

Short Communication

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Genetic evidence for the microphallid trematode *Maritrema graciosum* in the Northeast Pacific and an overview of its global distribution

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Abstract

The trematode *Maritrema graciosum* Nicoll, 1907 has a heteroxenous life cycle comprising a marine gastropod as its first intermediate host, a barnacle as its second intermediate host, and a bird as its definitive host. Globally, the species is widely distributed from subtropical waters of the South China Sea to arctic waters of the Barents Sea. In European waters (namely, western Europe and the Barents Sea), the complete life cycle of *M. graciosum* is known. Elsewhere in its range, however, its life cycle remains incompletely understood. Clarifying its distribution and resolving unknown host associations are essential for advancing a more holistic view of host–parasite dynamics and their ecological role in intertidal ecosystems. In this study, *M. graciosum* was detected using DNA-based assays in individuals of the barnacle *Balanus glandula* Darwin, 1854 that were collected from the rocky intertidal zone along the southern coast of British Columbia, Canada. This finding provides the first molecular evidence of this trematode species in the Pacific, complementing earlier records. The same molecular approach can be extended to identify region-specific hosts, such as the first intermediate (molluscan) host in the Pacific region, which remains unidentified.

Introduction

Maritrema graciosum Nicoll, 1907 (Digenea: Microphallidae) is a microphallid trematode with a heteroxenous life cycle involving intertidal invertebrates as first and second intermediate hosts and marine coastal birds as definitive hosts. First described from British shorebirds in the early 20th century (Nicoll 1907), the species has been known under various synonyms, including *Cercaria balani* Lebour, 1908, *Cercaria littorinae saxatilis* V Popiel, 1976, and *Maritrema arenaria* Hadley & Castle, 1940 (Deblock and Stam 1960; Deblock and Tran Van Ky 1966; Irwin et al. 1990). The species is primarily reported from the temperate waters of western Europe (Northeast Atlantic), where its life cycle has been characterised and where it has been the subject of detailed morphological, experimental, and, more recently, molecular work (Chuang 2021; Chuang et al 2025; Deblock and Tran Van Ky 1966; Hadley and Castle 1940; Irwin 1983; Irwin et al. 1990; Nicoll 1907; Popiel 1976a, b, 1978; Tkach et al 2003).

The life cycle of *M. graciosum* begins when the miracidia hatch from eggs in the intestine of its first intermediate host (a marine gastropod). Within the reproductive system of the gastropod, miracidia develop into sporocysts and undergo asexual reproduction to produce daughter sporocysts (Carrol et al. 1990; Popiel, 1978). In turn, daughter sporocysts give rise to clonal cercariae, which are ultimately released into the water column (Carrol et al. 1990). Although motile, cercariae exhibit limited swimming ability (Hadley and Castle, 1940; Williams et al. 1981). The free cercariae infect the second intermediate host (a barnacle). Cercariae enter the body of the barnacle via the cirri during feeding and encyst as metacercariae embedded in the intestinal wall (Irwin et al. 1990).

Metacercariae remain encysted until the second intermediate host (the barnacle) is ingested by the definitive host, which is typically a shorebird (Charadriiformes) or a waterfowl (Anseriformes), and less commonly a passerine (Passeriformes) or a grebe (Podicipediformes) (Ching 1978; Deblock and Rausch 1972; Hadley and Castle 1940; Nicoll 1907; Presswell et al. 2014). Within the digestive tract of the avian host, metacercariae excyst and develop into sexually mature adults. Subsequently, eggs are produced and, eventually, expelled into the marine environment through defecation, where they are ingested by the first intermediate (molluscan) host, completing the life cycle of *M. graciosum*.

Beyond the known hosts distributed in the Northeast Atlantic, *M. graciosum* has been reported from a wide range of other hosts and regions. Yet, knowledge about its global

distribution and life cycle remains fragmented, with many records based on morphology and incomplete identification of intermediate or definitive hosts. To synthesise this information, we consolidated published occurrence records to reconstruct the species' global distribution and identify regions where its life cycle remains unresolved. Further, genetic evidence in support of species identity remains geographically limited for such a widespread trematode species. In particular, all known genetic sequences of *M. graciosum* are from trematodes collected in the Northeast Atlantic (Tkach *et al.* 2003; Chuang 2021). Here, we employed genetic methods to confirm the taxonomic identity of metacercariae of *M. graciosum* recovered from barnacles in the Northeast Pacific (British Columbia), supporting previous identifications in this region (Ching 1978, 1991; MacLeod *et al.* 2021). Through this case study and a comprehensive review of published records, we aim to clarify the known distribution of this trematode and to assess the extent to which its life cycle and host associations have been described across its range.

Material and methods

Collection of animals

In December 2020, individuals of *Balanus glandula* Darwin, 1854 were collected from Point Grey Park (49°16'26.0"N; 123°09'52.2"W), along the southern shore of Burrard Inlet in Vancouver, British Columbia, Canada – a site previously reported to exhibit a high prevalence of metacercarial infection (*sensu* MacLeod *et al.* 2021). Collections targeted individuals of *B. glandula* inhabiting the intertidal zone. In the laboratory, somatic tissues of *B. glandula* were extracted using dissection forceps and examined under a light microscope for the presence of putative metacercariae of *Maritrema graciosum* (Figure 1). Infected tissues from two individuals were preserved in 95% ethanol and stored at –20 °C until extraction of deoxyribonucleic acid (DNA).

Genetic analysis

Genomic DNA was extracted from preserved trematode cyst tissues in *B. glandula* using the DNeasy Blood & Tissue Kit (QIAGEN)

following the manufacturer's protocols, with the exception that samples were eluted in a final volume of 50 µL. DNA concentrations were assessed using a Qubit fluorometer (Thermo Fisher Scientific).

Amplification via polymerase chain reaction (PCR) was used to target a partial region (~750 bp) of the 28S ribosomal RNA gene (28S) using the primers T16: 5'-GAG ACC GAT AGC GAA ACA AGT AC-3' and T30: 5'-TGT TAG ACT CCT TGG TCC GTG-3' (Harper and Saunders 2001). Additional amplification was used to target a partial region of the cytochrome c oxidase subunit I (COI) gene using the primers JB3: 5'-TTT TTT GGG CAT CCT GAG GTT TAT-3' and Trem.cox1.rnrl: 5'-AAT CAT GAT GCA AAA GGT A-3' (Bowles and McManus 1993; Král'ová *et al.* 2001).

PCRs were prepared with 12.5 µL FroggaBio 2×Taq, 0.6 µL of each forward and reverse primer, 3.55 µL of Bovine Serum Albumin (New England BioLabs), 2.55 µL of UltraPure DNase/RNase-Free Distilled Water (Thermo Fisher Scientific), and 5 µL of genomic DNA for a final volume of 25 µL. PCR conditions for samples prepared with the T16/T30 and JB3/Trem.cox1.rnrl primer sets included the polymerase activation of 94 °C for three minutes, followed by 40 cycles of denaturation and amplification at 94 °C for 30 seconds, 58 °C for 45 seconds, and 72 °C for one minute, then followed by a final elongation step of 72 °C for eight minutes. PCR amplification products were visualised on a 1.5% agarose gel stained with Red Safe (FroggaBio).

Non-purified PCR product was submitted to the Centre d'expertise et de services Génome Québec (Montréal, Canada) for Sanger sequencing. Sequences were obtained in both forward and reverse directions and edited using Geneious (version 11.0.4). A consensus sequence was formed by merging the forward and reverse sequences via *de novo* assembly.

Taxonomic assignment was carried out using BLASTn searches via the Basic Local Alignment Search Tool (BLAST) against the nucleotide database (nr/nt) of the National Center for Biotechnology Information (NCBI). Further taxonomic validation was carried out by placing our unknown sequences into a neighbour-joining tree with previously published *Maritrema* spp. sequences obtained from NCBI. This tree was constructed in Geneious v11.1.5 with 100 bootstrap replicates and all sequences trimmed to equal length (468 bp).

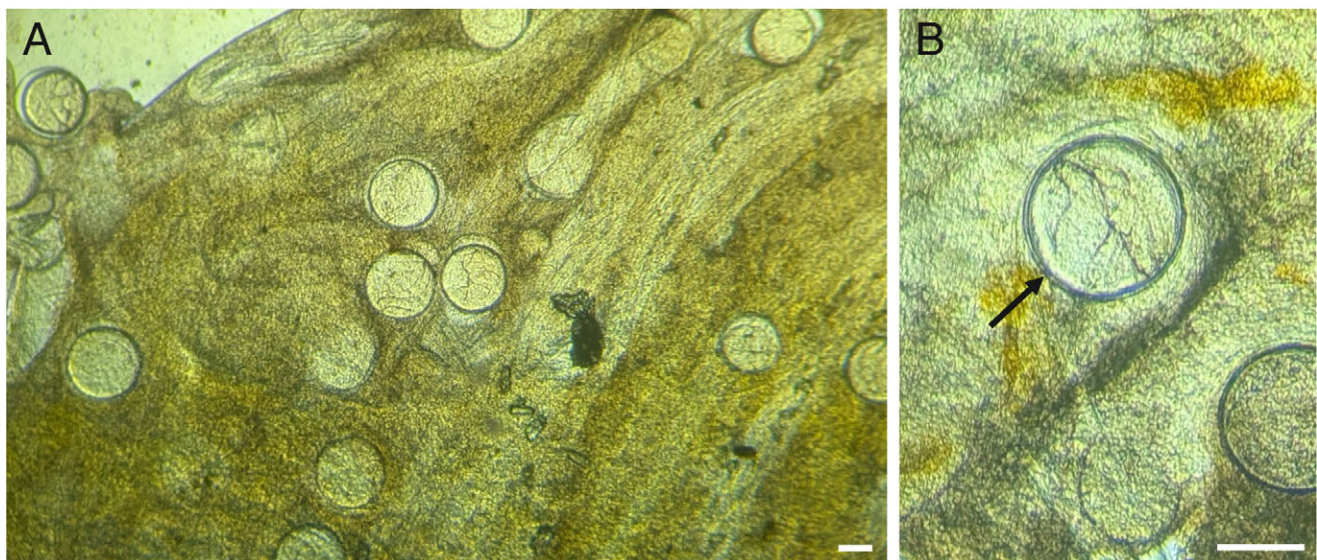


Figure 1. (A) Encysted metacercariae of *Maritrema graciosum* in the somatic tissue of *Balanus glandula*. (B) Close-up view of a metacercaria (black arrow). Scale bars represent 300 µm.

Results and discussion

In this study, we used molecular tools to support the identity of the trematode parasitising the barnacle *Balanus glandula* in British Columbia, Canada, as *Maritrema graciosum*. This represents the first molecular confirmation of *M. graciosum* outside western Europe. Amplification using T16 / T30 (28S) was successful in yielding parasite DNA sequences from two samples (Barcode of Life Data System [BOLD]; <https://doi.org/10.5883/DS-TREM>). These sequences were >99% identical to *Maritrema graciosum* (previously *Maritrema arenaria*) from Northern Ireland (NCBI Accession: AY220629.1; Tkach et al. 2003). Tree-based analysis also placed our sequences into a clade with this species with zero nucleotide differences (Figure 2). The next closest clade was *Maritrema prosthometra* Deblock & Heard, 1969, isolated from rodents in Florida, with 96% similarity (20 nucleotide differences; AY220631.1; Galaktionov et al. 2012; Tkach et al. 2003). In a separate analysis, sequences isolated from *Semibalanus balanoides* (Linnaeus 1767) collected from Scotland were identical to those from Northern Ireland (Chuang, 2021). We also obtained

sequences using the COI primers (JB3/Trem.cox1.rnrl) for the same specimens. However, these sequences had low similarity to publicly available reference sequences in BOLD and NCBI, and were not informative for inferring species identity. All DNA sequence data from both markers are publicly accessible in BOLD (<https://doi.org/10.5883/DS-TREM>) and NCBI (COI: PZ385258–PZ385259; 28S: PZ385594–PZ385595).

Based on published occurrence records derived from both morphological and molecular identifications, *M. graciosum* has a broad distribution spanning subtropical to arctic waters across both the Northern and Southern Hemispheres (Figure 3). In subtropical waters, the species has been detected in the South China Sea and the Gulf of Mexico (Figure 3). In temperate waters, records extend across western Europe, the Mediterranean Sea, British Columbia, the eastern United States, and New Zealand, while occurrences in subarctic waters span the Kuril Islands, the Bering Sea, and the Gulf of Alaska (Figure 3). At the northernmost extent of its range, the species has been documented in the arctic waters of the Barents Sea (Figure 3). One published record was excluded from this review.

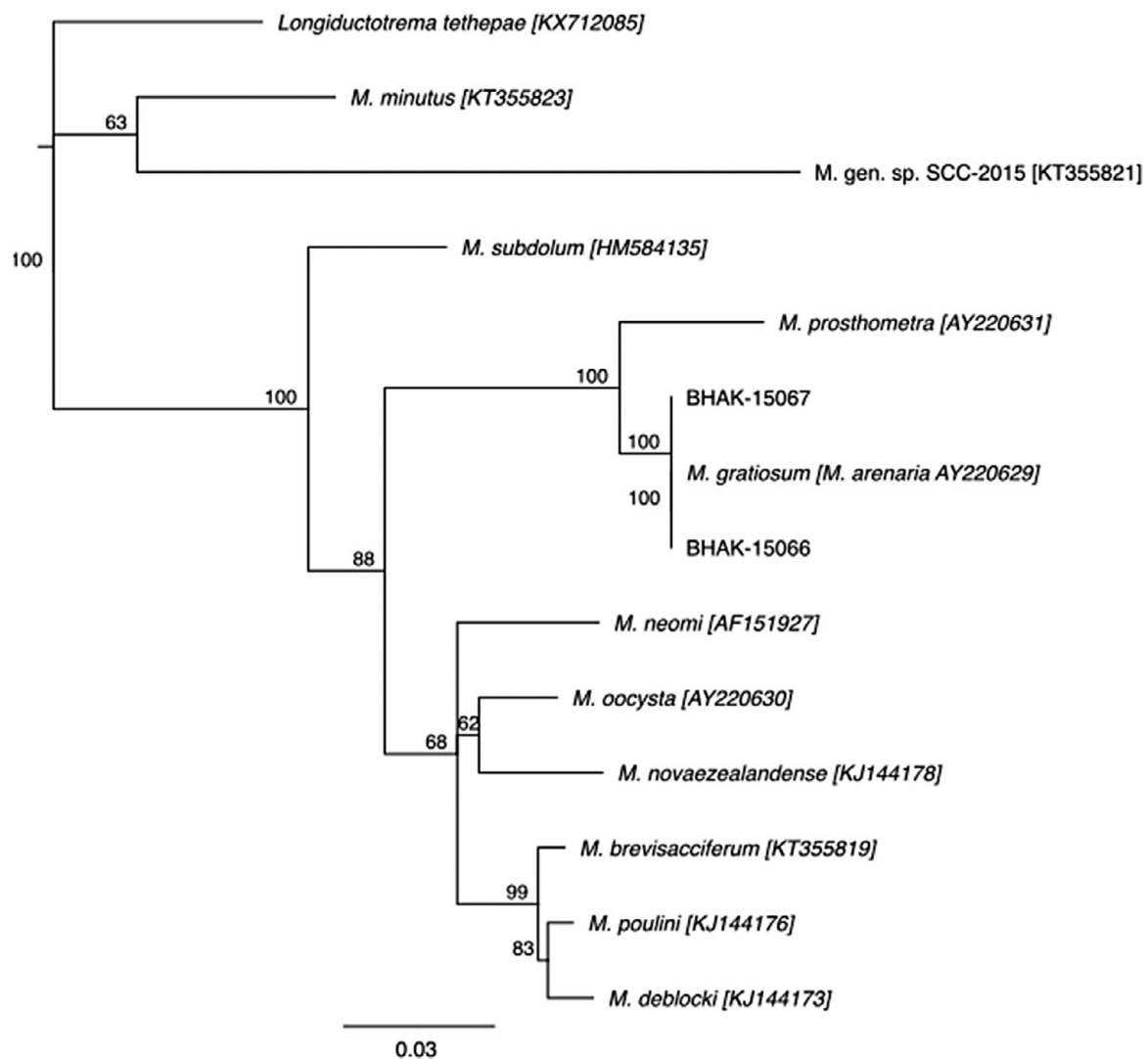


Figure 2. Neighbour-joining tree combining *Maritrema* sp. sequences from this study (BHAK-15066 and BHAK-15067) with previously published sequences of *Maritrema* spp. and *Microphallus* spp. (family Microphallidae) obtained from NCBI. Branch labels are consensus support (%), and tip labels for public records include the corresponding GenBank accession numbers.

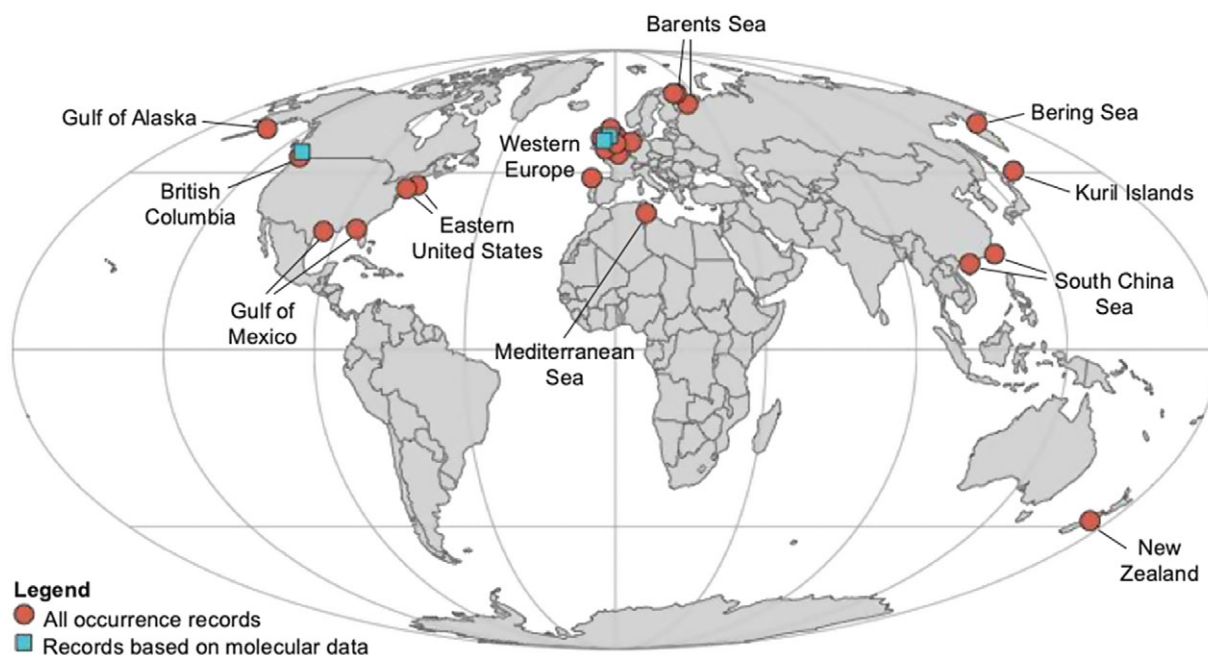


Figure 3. Global distribution of the microphallid trematode *Maritrema gratosum* based on occurrence records reported in the literature, spanning subtropical (South China Sea, Gulf of Mexico), temperate (western Europe, Mediterranean Sea, British Columbia, eastern United States, New Zealand), subarctic (Kuril Islands, Bering Sea, Gulf of Alaska), and arctic waters (Barents Sea). A subset of these records was supported by molecular data, including records from temperate waters in British Columbia (this study), Northern Ireland (Tkach et al 2003), and Scotland (Chuang 2021). Sources of distributional data are provided in Table 1.

Hassan et al. (2018) reported a trematode from marine gastropods in the Red Sea, attributing it to *Cercaria littorinae saxatilis* V (a synonym of *M. gratosum*; Irwin et al. 1990). Although the tail length was comparable to prior descriptions, the daughter sporocysts, cercariae, and stylets were substantially smaller than those reported for *M. gratosum*, and the ventral sucker was described as absent rather than present (Irwin et al. 1990; Popiel 1976b). These discrepancies suggest a likely misidentification.

A review of the literature indicates that only in western Europe and the Barents Sea is the complete life cycle of *M. gratosum* resolved, and the identity of associated hosts (or at least one complete set of hosts required to complete the life cycle) is known. From these regions, the hosts consisted of the periwinkle *Littorina saxatilis* (Olivi 1792) as its first intermediate host, *S. balanoides* and other barnacle species as its second intermediate host, and various species of birds as its definitive host (Table 1). In contrast, the life cycle of *M. gratosum* remains incompletely known in other parts of its range. In British Columbia and the eastern United States, only two of three hosts (namely, the second intermediate and definitive hosts) have been identified (Table 1). To date, only the second intermediate host has been identified from the Bering Sea, and only the definitive host from the South China Sea, the Gulf of Mexico, New Zealand, and the Gulf of Alaska (Table 1). Host identity was not specified for the records of *M. gratosum* from the Kuril Islands (Table 1). The molecular approach used in this study could be employed to identify new hosts (including paratenic hosts), which can contribute to a more complete understanding of the trematode's life cycle for regions outside of western Europe and the Barents Sea.

The survey of the literature revealed that *L. saxatilis* is the only first intermediate host species in which *M. gratosum* has been identified to date (Galaktionov and Bustnes 1999; Irwin 1983; McCarthy et al. 2002; Popiel 1976b), a host association further

supported by experimental transmission studies (Irwin et al. 1990). Beyond *B. glandula* and *S. balanoides*, known second intermediate host species include the barnacles *Austrominius modestus* (Darwin, 1854), *Chthamalus montagui* Southward, 1976, *Perforatus perforatus* (Bruguière, 1789), and *Semibalanus cariosus* (Pallas, 1788) (Ching 1978; Chuang 2021; Sari and Malek 2000; Williams et al. 1981). The most commonly reported definitive hosts are shorebirds, including sandpipers (genus *Calidris*), plovers (*Charadrius*), gulls (*Chroicocephalus* and *Larus*), oystercatchers (*Haematopus*), turnstones (*Arenaria*), terns (*Sterna*), curlews (*Numenius*), godwits (*Limosa*), and greenshanks (*Tringa*) (Table 2). Infections have also been described in several waterfowl – a goldeneye (*Bucephala*), a scoter (*Melanitta*), and an eider (*Somateria*) – as well as in a passerine (*Corvus*) and a grebe (*Podiceps*) (Table 2). In addition to these obligate hosts, the marine gastropod *Nucella lapillus* (Linnaeus 1758) has been identified as a paratenic host of *M. gratosum* (McCarthy et al. 1999). This gastropod is a predator of the barnacle *S. balanoides*, a known second intermediate host (Irwin et al. 1990), and is itself consumed by birds, the definitive hosts (McCarthy et al. 1999). The detection of metacercarial cysts in *N. lapillus* suggests that infection occurs through the consumption of infected barnacles. Paratenic hosts are thought to augment and maintain transmission between the second intermediate and avian hosts (Marcogliese 2007).

Building on earlier records from British Columbia (Ching 1978, 1991; MacLeod et al. 2021), this study provides the first molecular evidence confirming the presence of *M. gratosum* in this region and extends its molecularly confirmed range from the North Atlantic to the Northeast Pacific. Confirming the presence of this trematode species in the barnacle *B. glandula* will pave the way to a better understanding of host–parasite dynamics, given suggestions that *M. gratosum* may reduce the survival of its second intermediate host and influence the barnacle's distribution on intertidal rocky shores (Carroll et al. 1990; Mouritsen and Poulin 2002). Yet, a

Table 1. Hosts in the life cycle of the microphallid trematode *Maritrema graciosum* that have been identified in the literature. Names of host species were updated to their current valid nomenclature. Names of regions correspond to those shown in Figure 3

Climate and region	Identified hosts in the life cycle of the trematode	Host species	Sources
Subtropical			
South China Sea	Definitive host	<i>Arenaria interpres interpres</i> , <i>Charadrius alexandrinus dealbatus</i> , <i>Tringa nebularia</i>	Fischthal and Kuntz (1976), Ke (1976)
Gulf of Mexico	Definitive host	<i>Limnodromus griseus</i> , <i>Numenius americanus</i>	Loftin (1960), Dronen and Badley (1979)
Temperate			
Western Europe	First intermediate host	<i>Littorina saxatilis</i>	Popiel (1976b, 1978), Irwin (1983), Irwin et al. (1990), McCarthy et al. (2002)
	Second intermediate host	<i>Austrominius modestus</i> , <i>Chthamalus montagui</i> , <i>Perforatus perforatus</i> , <i>Semibalanus balanoides</i>	Lebour (1908, 1911), Deblock and Stam (1960), Deblock and Tran Van Ky (1966), Popiel (1976a), Irwin and Irwin (1980), Williams et al. (1981), Mitchell and Dessi (1984), Carrol et al. (1990), Irwin et al. (1990), Sari and Malek (2000), Chuang (2021) and references therein, Chuang et al. (2025)
	Paratenic host	<i>Nucella lapillus</i>	McCarthy et al. (1999)
	Definitive host	<i>Calidris alpina</i> , <i>Charadrius hiaticula</i> , <i>Chroicocephalus ridibundus</i> , <i>Haematopus ostralegus</i> , <i>Larus argentatus</i> , <i>L. michahellis</i> , <i>Limosa lapponica lapponica</i> , <i>Melanitta nigra</i> , <i>Somateria mollissima</i>	Nicoll (1907), Baylis (1939), Rankin (1939), Deblock and Tran Van Ky (1966), Dawes (1968), Irwin and Prentice (1976), Borgsteede et al. (2005), Sanmartín et al. (2005), Álvarez et al. (2006); also see Kuklin (2022), Kuklin and Kuklina (2021)
Mediterranean Sea	Definitive host	<i>Calidris minuta</i>	Balozet and Callot (1939); also see Dawes (1968) and Deblock and Canaris (1992)
British Columbia	Second intermediate host	<i>Balanus glandula</i>	Ching (1978, 1991), this study; also see MacLeod et al. (2021)
	Definitive host	<i>Bucephala islandica</i>	Ching (1978); also see Irwin and Irwin (1980), Kuklin (2022)
Eastern United States	Second intermediate host	<i>Semibalanus balanoides</i>	Hadley and Castle (1940), Arvy and Nigrelli (1969) ^a
	Definitive host	<i>Arenaria interpres morinella</i>	Hadley and Castle (1940); also see Irwin and Irwin (1980), Kuklin (2022)
New Zealand	Definitive host	<i>Haematopus finschi</i> , <i>Limosa lapponica baueri</i>	Allison (2000, 2001), Presswell et al. (2014); also see Kuklin (2022)
Subarctic			
Kuril Islands	Not specified	No data	Deblock and Tran Van Ky (1966) and references therein
Bering Sea	Second intermediate host	<i>Semibalanus cariosus</i>	Ching (1978) and references therein
Gulf of Alaska	Definitive host	<i>Corvus brachyrhynchos</i> , <i>Haematopus bachmani</i>	Deblock and Rausch (1972)
Eastern Siberia ^b	Not specified	No data	Deblock and Tran Van Ky (1966), Deblock and Rausch (1972) and references therein
Arctic			
Barents Sea	First intermediate host	<i>Littorina saxatilis</i>	Galaktionov and Bustnes (1999)
	Second intermediate host	<i>Semibalanus balanoides</i>	Ouspenskaia (1960), Deblock and Tran Van Ky (1966), Ching (1978) and references therein
	Definitive host	<i>Larus argentatus</i> , <i>L. hyperboreus</i> , <i>L. marinus</i>	Kuklin (2015, 2022), Kuklin and Kuklina (2021)

^aMost likely to be *M. graciosum*.

^bMost likely referring to localities on the Kuril Islands or in the Bering Sea.

Table 2. Documented definitive hosts of the microphallid trematode *Maritrema gratosum*. Names of host species were updated to their current valid nomenclature. Names of regions correspond to those shown in Figure 3

Host species	Common name	Regions	Sources
Charadriiformes			
<i>Arenaria interpres interpres</i> (Linnaeus, 1758)	Ruddy turnstone	South China Sea	Fischthal and Kuntz (1976)
<i>Arenaria interpres morinella</i> (Linnaeus, 1766)	Ruddy turnstone	Eastern United States	Hadley and Castle (1940), Irwin and Irwin (1980), Kuklin (2022)
<i>Calidris alpina</i> (Linnaeus, 1758)	Dunlin	Western Europe	Nicoll (1907), Baylis (1939), Rankin (1939), Deblock and Tran Van Ky (1966), Dawes (1968)
<i>Calidris minuta</i> (Leisler, 1812)	Little stint	Mediterranean Sea	Balozet and Callot (1939); also see Dawes (1968) and Deblock and Canaris (1992)
<i>Charadrius alexandrinus dealbatus</i> (Swinhoe, 1870)	White-faced Kentish plover	South China Sea	Ke (1976)
<i>Charadrius hiaticula</i> (Linnaeus, 1758)	Common ringed plover	Western Europe	Nicoll (1907), Rankin (1939), Deblock and Tran Van Ky (1966), Dawes (1968)
<i>Chroicocephalus ridibundus</i> (Linnaeus, 1766)	Black-headed gull	Western Europe	Nicoll (1907), Rankin (1939), Dawes (1968), Irwin and Prentice (1976), Irwin and Irwin (1980), Sonin (1985)
<i>Haematopus finschi</i> G.H. Martens, 1897	South Island oystercatcher	New Zealand	Allison (2000, 2001), Presswell et al. (2014), Kuklin (2022)
<i>Haematopus bachmani</i> Audubon, 1838	Black oystercatcher	Gulf of Alaska	Deblock and Rausch (1972)
<i>Haematopus ostralegus</i> Linnaeus, 1758	Eurasian oystercatcher	Western Europe	Nicoll (1907), Rankin (1939), Dawes (1968), Irwin and Irwin (1980)
<i>Larus argentatus</i> Pontoppidan, 1763	Herring gull	Western Europe, Barents Sea	Irwin and Prentice (1976), Irwin and Irwin (1980), Sonin (1985), Kuklin (2022), Kuklin and Kuklina (2021)
<i>Larus canus</i> Linnaeus, 1758	Common gull	Not specified	Sonin (1985)
<i>Larus hyperboreus</i> Gunnerus, 1767	Glaucous gull	Barents Sea	Kuklin (2015, 2022)
<i>Larus marinus</i> Linnaeus, 1758	Great black-backed gull	Barents Sea	Kuklin (2022)
<i>Larus michahellis</i> Naumann, 1840	Yellow-legged gull	Western Europe	Sanmartín et al. (2005), Álvarez et al. (2006), Kuklin (2022)
<i>Limnodromus griseus</i> (J.F. Gmelin, 1789)	Short-billed dowitcher	Gulf of Mexico	Loftin (1960)
<i>Limosa lapponica baueri</i> Naumann, 1836	Bar-tailed godwit	New Zealand	Allison (2001), Presswell et al. (2014), Kuklin (2022)
<i>Limosa lapponica lapponica</i> (Linnaeus, 1758)	Bar-tailed godwit	Western Europe	Baylis (1939), Deblock and Tran Van Ky (1966), Dawes (1968)
<i>Numenius americanus</i> Bechstein, 1812	Long-billed curlew	Gulf of Mexico	Dronen and Badley (1979)
<i>Sterna paradisaea</i> Pontoppidan, 1763	Arctic tern	Not specified	Sonin (1985)
<i>Tringa nebularia</i> (Gunnerus, 1767)	Common greenshank	South China Sea	Ke (1976)
Anseriformes			
<i>Bucephala islandica</i> (J.F. Gmelin, 1789)	Barrow's goldeneye	British Columbia	Ching (1978), Irwin and Irwin (1980), Kuklin (2022)
<i>Melanitta nigra</i> (Linnaeus, 1758)	Common scoter	Western Europe	Baylis (1939), Deblock and Tran Van Ky (1966), Dawes (1968)
<i>Somateria mollissima</i> (Linnaeus, 1758)	Common eider	Western Europe	Borgsteede et al. (2005), Kuklin (2022)
Passeriformes			
<i>Corvus brachyrhynchos</i> Brehm, 1822	American crow	Gulf of Alaska	Deblock and Rausch (1972)
Podicipediformes			
<i>Podiceps grisegena</i> (Boddaert, 1783)	Red-necked grebe	Not specified	Sonin (1985)

substantial gap remains in our understanding of the regional life cycle in British Columbia, most notably, the identity of the first intermediate host (MacLeod et al. 2021). A similar gap was highlighted for the eastern United States, where the first intermediate host of *M. gratosum* likewise remains unidentified (Hadley and Castle 1940). Future work to fully characterise these regional life cycles and assess the effects of *M. gratosum* on host species can help clarify the broader ecological consequences of trematode infection for marine ecosystems (Bennett et al. 2023).

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Competing interests. The authors declare that they have no competing interests.

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